



Coimisiún na Scrúduithe Stáit
State Examinations Commission

Leaving Certificate Examination 2024

Applied Mathematics

Mathematical Modelling Project
Reporting Booklet

Higher and Ordinary Level

100 marks

Examination Number

132334

Date of Birth

03/04/2006

Information for Candidates, Teachers and Schools

Completion of the report

- Your report on your mathematical modelling project must be completed in this booklet, using the prescribed structure. The report must be completed and handed over to your class teacher by **02 February 2024**. The report will be submitted through the SEC student portal.
- The State Examinations Commission will provide further information in due course, in the relevant circular, in relation to the appropriate file name to use and how your school will upload the digital coursework to the school portal.
- It is your responsibility to ensure that all electronic material submitted is free from viruses etc., so that the file can be opened for assessment.
- **Examiners will only mark work that is submitted as outlined in this booklet.**
- Your report must not exceed 900 words (excluding references, equations, diagrams, graphs, etc.). Your report must not include more than 20 images. An image can be any relevant table, graph, chart, diagram or photograph. The total file size of your digital report including all embedded images must not exceed 100 MB. Videos must not be included in your report.
- You must not change the structure or format of the booklet in any way and should adhere to the following formatting guidelines:
 - The text should be in Arial, font size 12.
 - You may make use of text editing features such as italics, bullets, etc.
 - Document margins have been set and should not be changed. The text margins for each A4 page are:
 - left margin: 20 mm
 - right margin: 20 mm
 - top margin: 20 mm
 - bottom margin: 35 mm
- You are not allowed to change these settings or otherwise circumvent these restrictions, which are set to facilitate online marking. You are also not allowed to attach or include links to any other material (other than references).
- When referring in the body of the report to any specific image, the image must be properly labelled (Figure 1, Figure 2, etc.). Images should not be used as a means to include additional text. It is advisable not to use images where a person or persons in these images may be identifiable.
- Penalties may apply where the overall word count or image count or file size is exceeded.

- You must reference and acknowledge all research sources used, such as: publications – including books, professional journals and government reports; online sources and other types of media; any material generated using artificial intelligence (AI) software or applications; specialist organisations and relevant individuals. To include such material without properly referencing the source will be considered plagiarism. In addition, the copying from, or reproduction of, material from such sources may also be considered plagiarism.
- Your project must be backed up by the school on a secure encrypted storage system until the State Examinations Commission is satisfied that the projects have been received and are accessible. This will require material being retained until the examination process is complete, including until after the appeal process is completed.

Authentication

- The mathematical modelling project and report must be your own individual work – authenticated by yourself, by your teacher and by your school management authority. Authentication is an important part of how the State Examinations Commission ensures fairness to everybody in the assessment of coursework.
- While you may carry out background research relevant to the brief on your own and/or at home, all other parts of your project and your report must be completed under the supervision of your teacher in accordance with the conditions set down by the State Examinations Commission as outlined in Circulars S69/04 and S68/08, which are available at www.examinations.ie.
- Any case of suspected copying, plagiarism (which includes the use of AI software), improper assistance, or procurement of work prepared by another party will be thoroughly investigated. These actions are breaches of examination rules and attract the penalties described in the *Rules and Programme for Secondary Schools*. The penalties include: loss of the marks for the project, loss of the subject, loss of the entire examination in all subjects, or being debarred from the Certificate Examinations in subsequent years. There may be serious consequences for any person who provides you with inappropriate assistance, as this is an offence under the Education Act 1998.
- Your teacher must supervise your completion of both the project and the report. If your teacher cannot confirm that the project and report are your own work, and that you carried out the project and completed the report under your teacher's supervision, the State Examinations Commission will not accept your report for assessment. In that case, you will forfeit the marks for this component of the examination. Teachers and the authorities of schools are familiar with the detailed requirements to ensure that practical and project work is valid for examination purposes. You should comply fully with all requests that are made by the teacher and the school in order to enable authentication of your work.
- Responsibility for complying with examination requirements rests with you, the candidate. If the requirements are not followed, your teacher and school will have no choice but to bring this to the attention of the State Examinations Commission.

Report Structure and Mark Allocations

Section	Indicative Content	Marks
<i>Introduction and Research</i>	• Background research on brief • Identify specific problem(s) to be modelled • Research specific problem(s) • Identify relevant variables • Present relevant data • Provide citations and references	20
<i>The Modelling Process</i>	• Explain and justify model and assumptions • Compute solutions • Present solutions using appropriate mathematical and graphical representations • Analysis of solution(s) – sensitivity to changes in assumptions; comparison with other solutions or real-world data • Iterative process	50
<i>Interpretation of Results</i>	• Interpretation of solution(s) in real-world context • Conclusions and reflections	15
<i>Communication and Innovation</i>	This is not a distinct section of the report. • Innovative and creative approaches • Overall coherence	15

Mathematical Modelling Project

Type into the space below the title of your mathematical modelling project.

Investigation and modelling of the exit velocity of a roller coaster going down a slope and into a curve

Word Count

Do not complete the following table until after you have finished your report.

<i>Section</i>	Number of words
<i>Introduction and Research</i>	196
<i>The Modelling Process</i>	468
<i>Interpretation of Results</i>	195
<i>Total</i>	859

In my applied maths project, I will be conducting mathematical modelling of a roller coaster as it goes through a section of a roller coaster track. The section of the track will be comprised of a steep hill joining into a sector of a circle that joins the ground.

My research consisted of initially looking for all the possible modellable forces that could slow down a rollercoaster. I have modelled 2 of these in my iterations.

To start the modelling, I broke my track down into 2 main parts, the hill and the curve. The hill has a slope of $\frac{-4}{3}$ and a length of 50m. The hill is tangential to the curve, and the point of contact is 6m above the ground.

ITERATION 1: This will consist of basic conservation of energy calculations. Ignoring all other energy losses.

ITERATION 2: I will then include friction in my calculations for the linear and circular part of my track.

ITERATION 3: This iteration will incorporate drag forces into the calculations.

CONSTANTS:

- Coefficient of friction - $\mu = 0.1$
- Mass of roller coaster cart – $m = 300kg$
- Acceleration due to gravity – $g = -9.81ms^{-2}$
- Air density – $\rho = 1.225kgm^{-3}$
- Cross sectional area – $A = 1.68m^2$
- Drag coefficient – $C_d = 0.2$

LINKS:

- **Cross section area:** <https://rideentertainment.com/roller-coasters-coaster-vehicles/>
- **Air density:** <https://www.toppr.com/guides/physics/fundamentals/density-of-air-how-to-calculate-air-density/>
- **Drag coefficient:** https://www.engineeringtoolbox.com/drag-coefficient-d_627.html
- **Coefficient of friction:** https://www.engineeringtoolbox.com/friction-coefficients-d_778.html

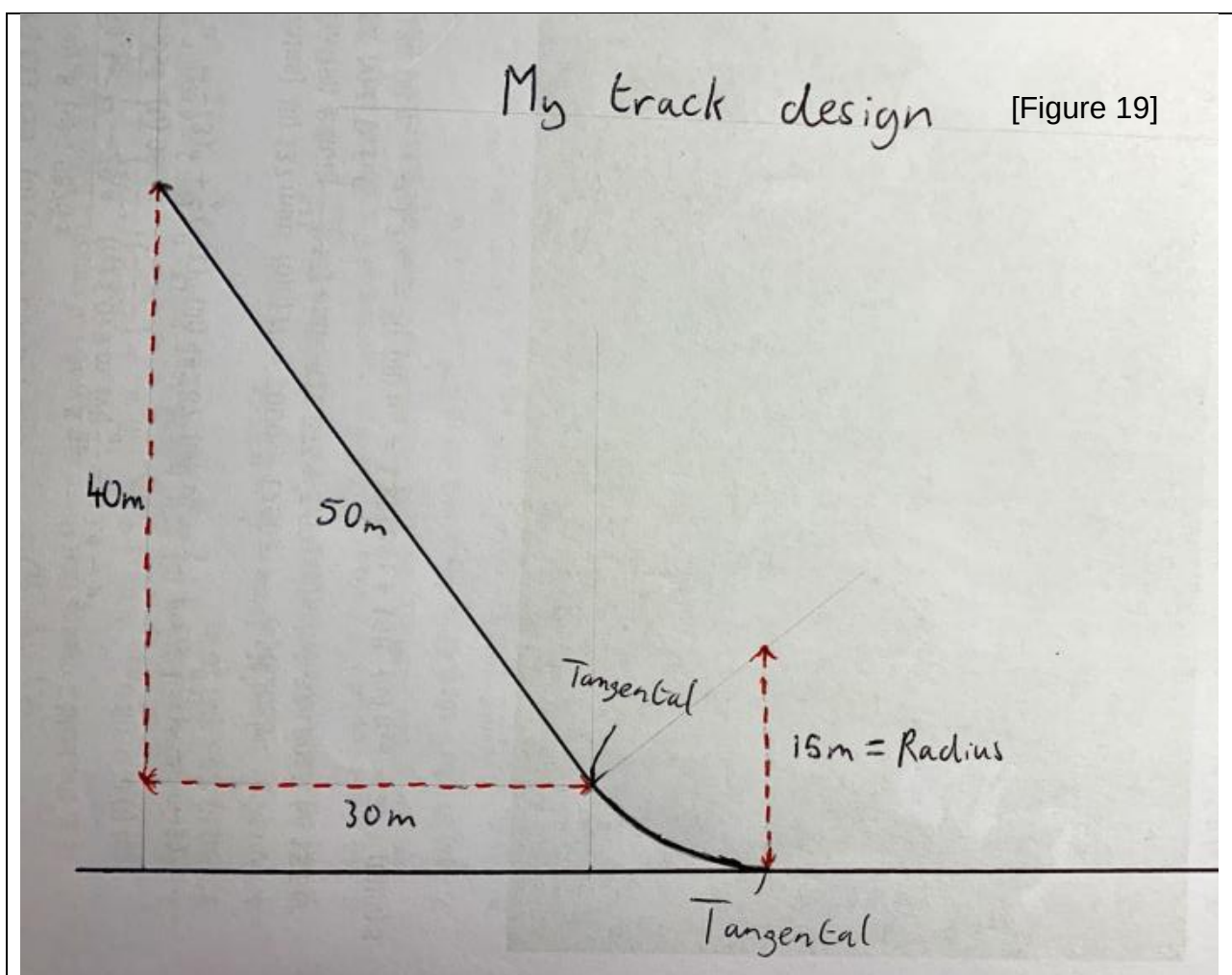
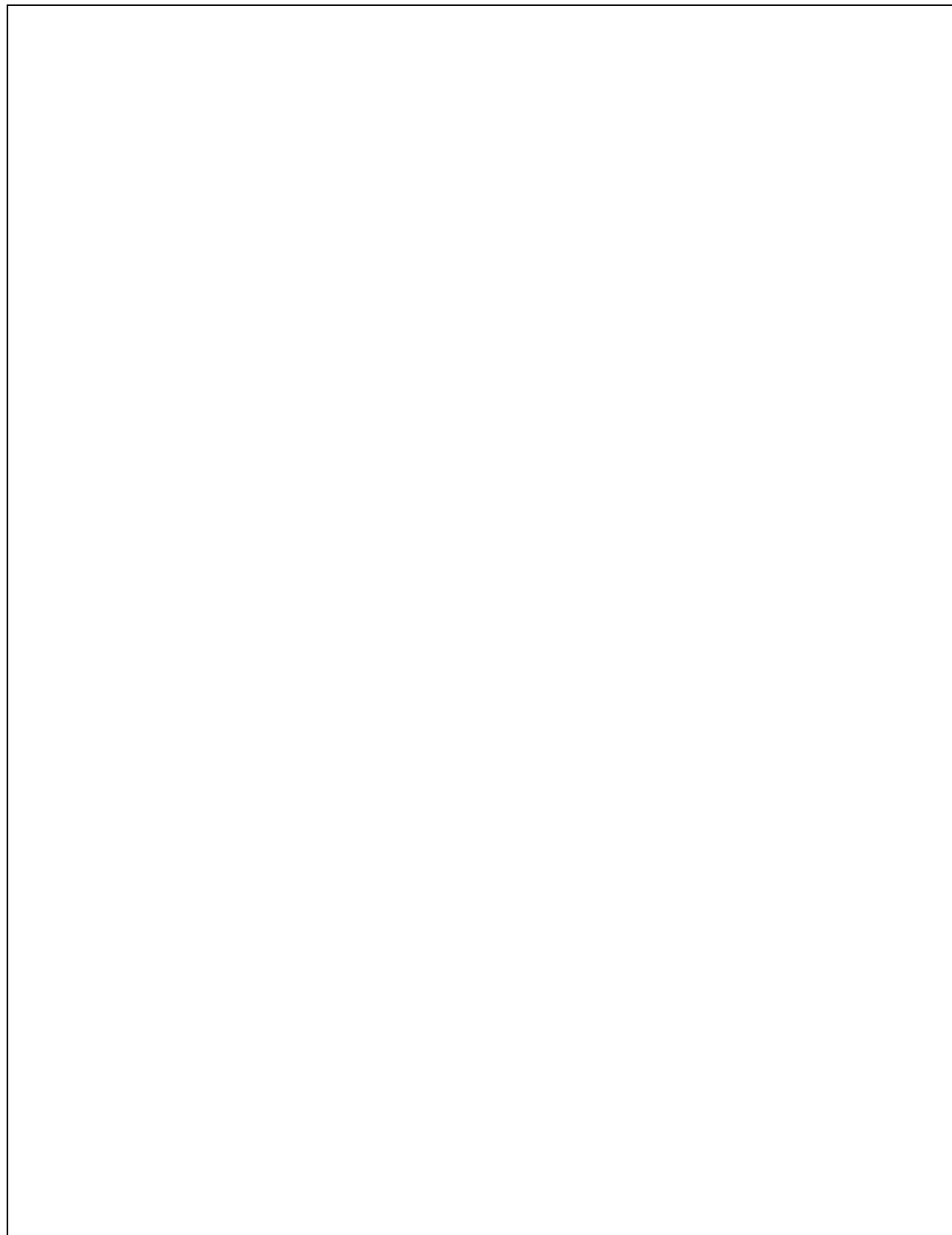
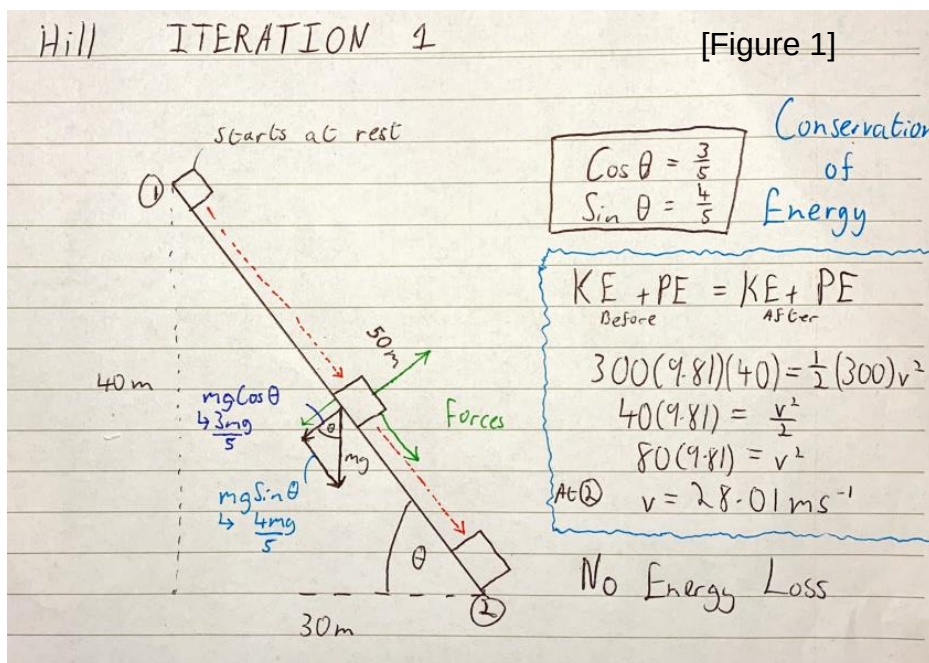


Figure 19 shows my track design.



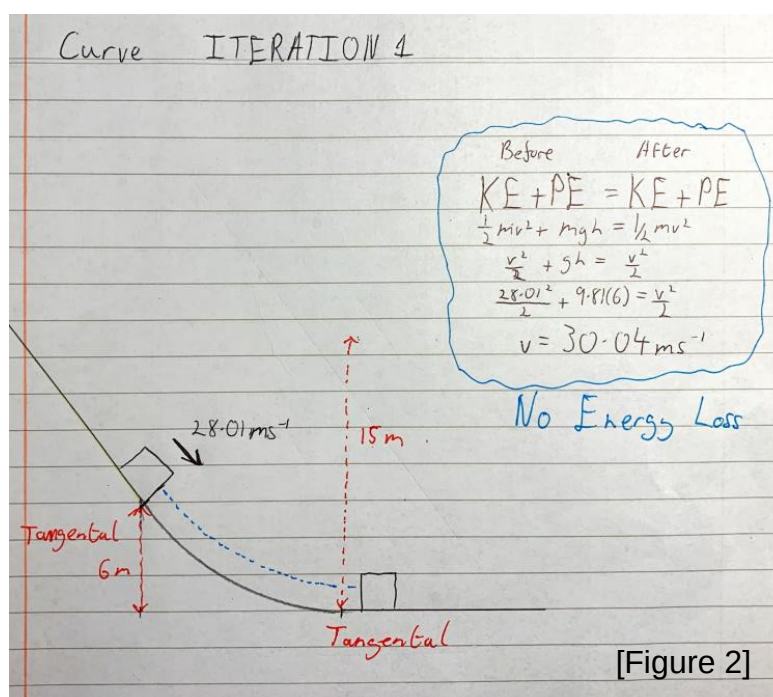
ITERATION 1

For my first Iteration I started with basic conservation of energy equations to find the energy at a certain height: $PE_{\text{Before}} + KE_{\text{Before}} = PE_{\text{After}} + KE_{\text{After}}$.



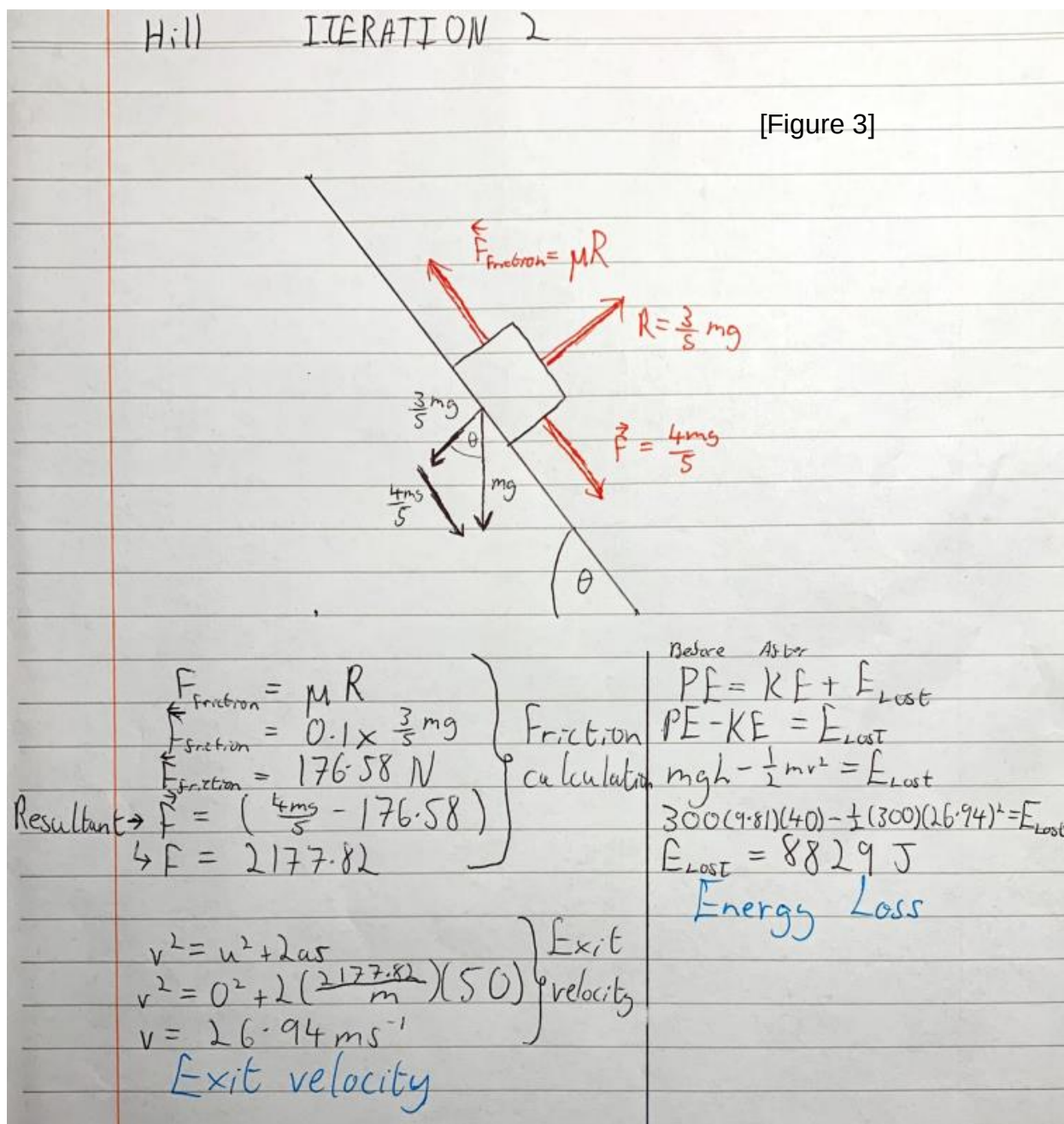
We see in figure 1 my use of the principal of conservation of energy to find the exit velocity from the hill of 28.01m/s.

We see here in figure 2 my use of the same energy formula to find the exit velocity from the curve of 30.04m/s.



ITERATION 2

In this iteration I incorporated friction force into my calculation by using the formula $F_{\text{friction}} = \mu R$. For this section of the track, the normal force $R = mg \cos \theta = \frac{3mg}{5}$. It was simpler for me to use the formula $v^2 = u^2 + 2as$ where a was my resultant force divided by mass, and s was my distance 50m. This gave me an exit velocity of 26.94m/s and an energy loss of 8829J.



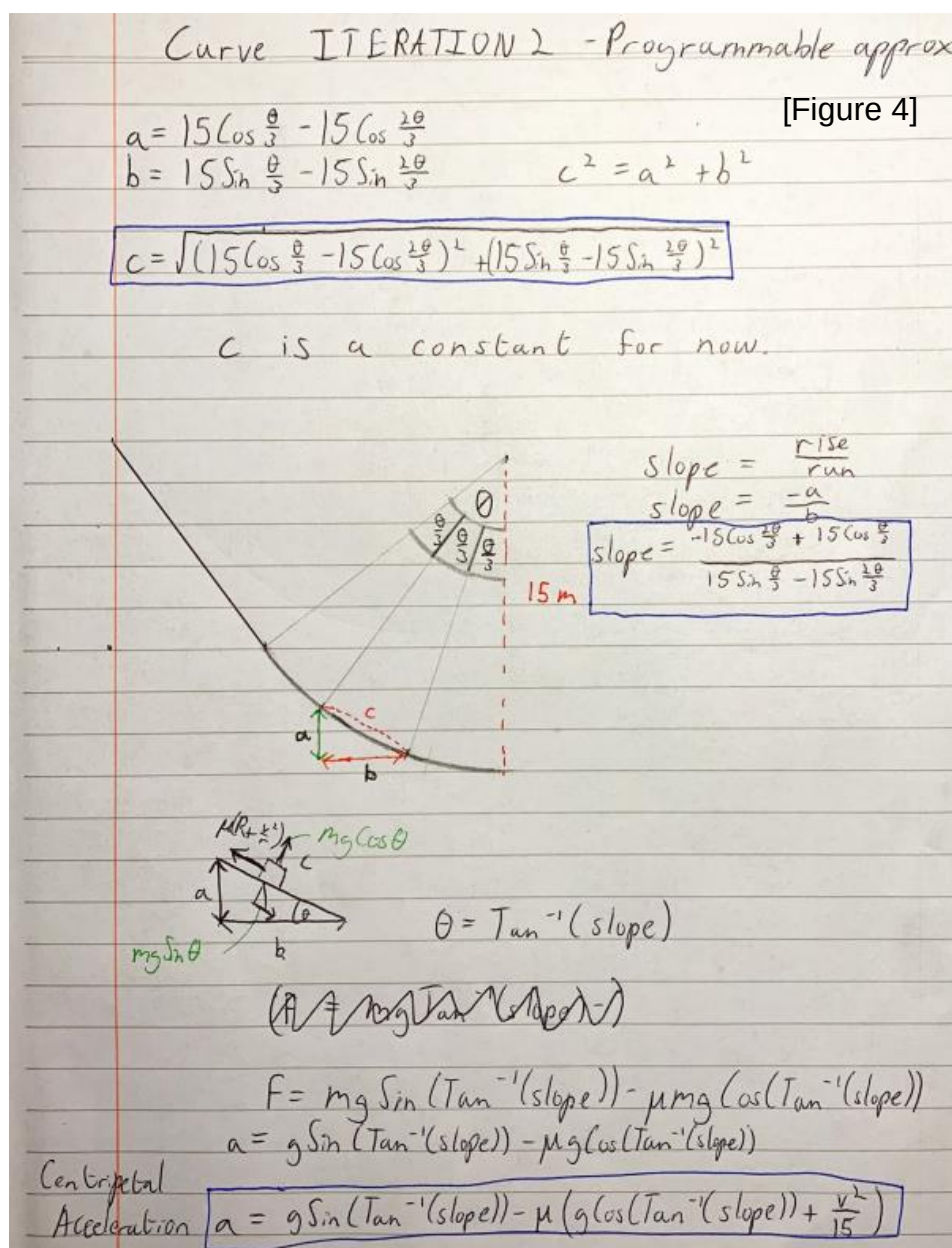
For the curve section of this iteration, I decided to use an approximation which breaks down the curve into sections and treats each section like a linear slope, similar to the hill section of the track. For each section I find the slope and then resolve the forces on that slope. The forces for any given section will look like this: $R = mg \cos(\tan^{-1}(\text{slope})) + \frac{mv^2}{r}$,

$$F_{\text{friction}} = \mu R, \vec{F} = mg \sin(\tan^{-1}(\text{slope})),$$

Formulas:

- $v^2 = u^2 + 2as$
- $a^2 + b^2 = c^2$
- $\text{slope} = \frac{\text{rise}}{\text{run}}$

The diagram and workings shown in this figure break up the curve into 3 sections rather than 100 to show how I came up with my approximation formula.



To make my approximation accurate I made a recursive function to express velocity. I then ran into a problem where the angle of the slope wasn't changing with each section of the curve, so then I made a nested function $T_n = \frac{\theta(\alpha+1)}{\alpha} - \frac{\theta n}{\alpha}$, where alpha is the degree of accuracy between 1 and infinite, and theta is constant. I coded a basic program to run through the 100 sections of the curve and found an exit velocity of 27.21m/s. This gave me a total energy loss of 24320.39J.

[Figure 5]

Curve ITERATION 2 - Manual recursive

Recursive function where v^2 gets subbed in for u^2 on the next slope section.

$$v^2 = u^2 + 2as$$

$$v^2 = u^2 + 2(g \sin(\tan^{-1}(\text{slope})) - \mu(g \cos(\tan^{-1}(\text{slope})) + \frac{u^2}{15}))(c)$$

TO MAKE IT DYNAMIC

Slope needs to change for each section, there for angle needs to be variable in the slope equation.

$$T_n = \frac{\theta(\alpha+1)}{\alpha} - \frac{\theta n}{\alpha} \quad 1 < \alpha < \infty$$

$\theta = \text{Constant}$, $n = \text{section of slope}$, $\alpha = \text{degree of accuracy}$

$$\text{slope} = \frac{15 \cos(T_n) - 15 \cos(T_{n+1})}{15 \sin(T_n) - 15 \sin(T_{n+1})}$$

$c = \text{Constant}$ but it has to reflect the degree of accuracy so it has to use the T_n function.

$$c = \sqrt{(15 \cos(T_1) - 15 \cos(T_2))^2 + (15 \sin(T_1) - 15 \sin(T_2))^2}$$

$$v = \left[u^2 + 2(g \sin(\tan^{-1}(\text{slope})) - \mu(g \cos(\tan^{-1}(\text{slope})) + \frac{u^2}{15}))(c) \right]$$

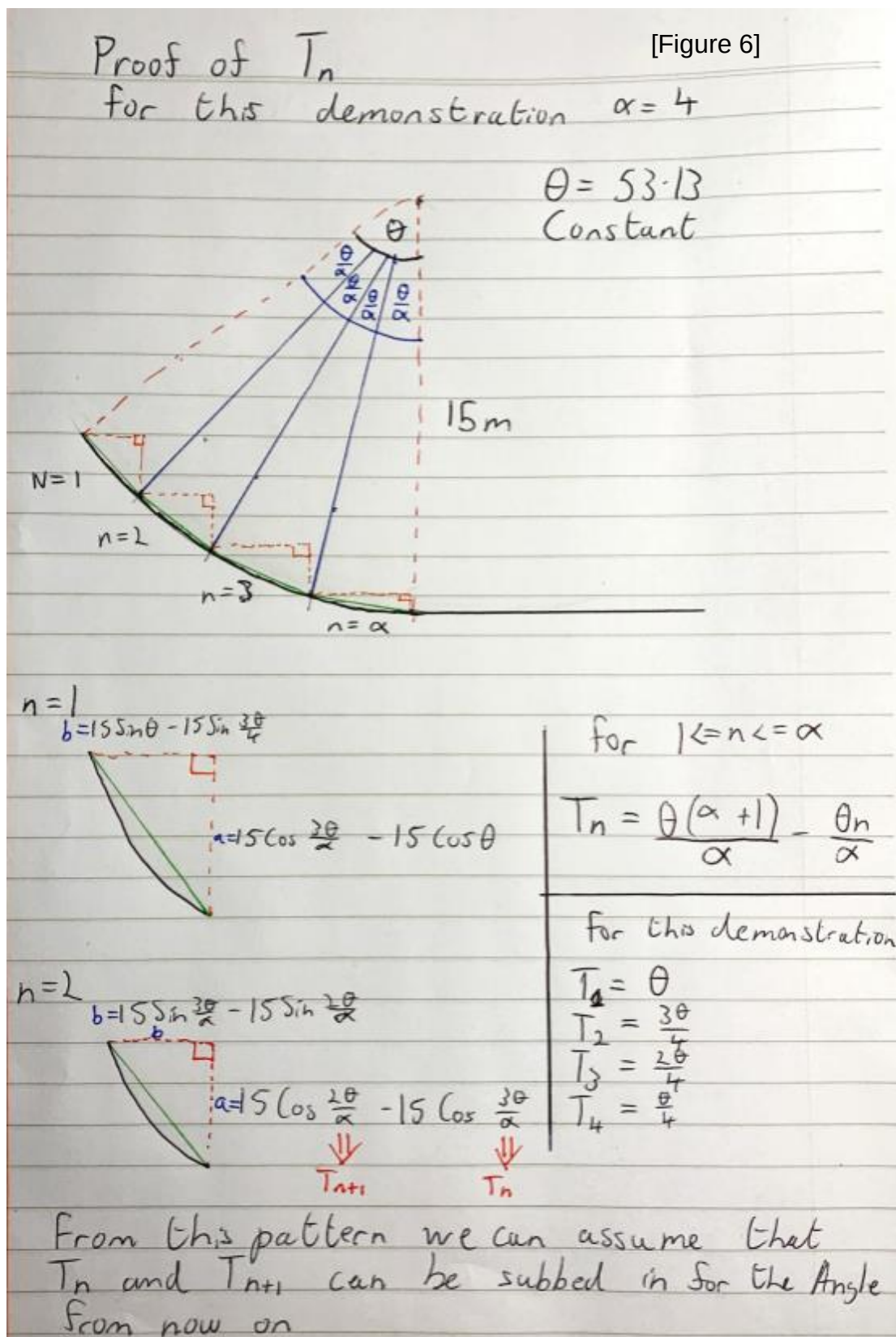
Exit velocity = 27.21 m/s

Loss in energy = $mgh - \frac{1}{2}mv^2$

$$= 300(9.81)(46) - \frac{1}{2}(300)(27.21)^2$$

$$= 24320.39 \text{ Joules}$$

As I mentioned previously, I created a nested function T_n to express the angle of the section with respect to n . T_n is a function of n . The result of the function is the angle for the portion of the circle n , as shown in my diagram demonstration below. This function allows the velocity function to become dynamic (see figure 6).



For my third iteration I factored drag into my calculations. Using the Energy formulas, I was able to calculate the energy loss for the hill section.

Formulas:

- $F_{\text{drag}} = \frac{1}{2} \rho v^2 C_d A$

[Figure 7]

Iteration 3 Hill section

$a = v \frac{dv}{ds}$

$$F = 300 \left(\frac{4}{5} (9.81) - 0.1 \left(\frac{3}{5} \right) (9.81) \right) - 0.2058 v^2$$

$$ma = 2177.82 - 0.2058 v^2$$

$$m v \frac{dv}{ds} = 2177.82 - 0.2058 v^2$$

$$m v \frac{dv}{ds} = p - kv^2$$

$$p = 2177.82$$

$$k = 0.2058$$

$$m = 300$$

$$\frac{m v}{p - kv^2} dv = ds$$

$$m \int \frac{v}{p - kv^2} dv = \int ds$$
~~$$m \int \frac{v}{u} dv = s + c$$~~

$$m \int \frac{\left(\frac{-1}{2k} \right)}{u} du = s + c$$

$$-\frac{m}{2k} \int \frac{1}{u} du = s + c$$

$$-\frac{m}{2k} \ln|u| = s + c$$

$$-\frac{m}{2k} \ln|p - kv^2| = s + c$$

$$-\frac{m}{2k} \ln|p - ku^2| = 0 + c$$

$$s = \frac{m}{2k} \ln|p - ku^2| - \frac{m}{2k} \ln|p - kv^2|$$

$$s = \frac{m}{2k} \ln \left| \frac{p - ku^2}{p - kv^2} \right|$$

$$\ln \left| \frac{p - ku^2}{p - kv^2} \right| = \frac{2ks}{m}$$

$$e^{\frac{2ks}{m}} = \frac{p - ku^2}{p - kv^2}$$

Let $u = p - kv^2$

$$\frac{du}{dv} = -2kv$$

$$\frac{-1}{2k} du = v dv$$

Initial Conditions
when $s=0$ $v=u$

$$e^{\frac{2ks}{m}} = \frac{p - ku^2}{p - kv^2}$$

$$(p - kv^2) e^{\frac{2ks}{m}} = p - ku^2$$

$$p e^{\frac{2ks}{m}} - kv^2 e^{\frac{2ks}{m}} = p - ku^2$$

$$v^2 = \frac{p e^{\frac{2ks}{m}} - p + ku^2}{k e^{\frac{2ks}{m}}}$$

$$v = \sqrt{\frac{p e^{\frac{2ks}{m}} - p + ku^2}{k e^{\frac{2ks}{m}}}}$$

When I apply this formula I created to my situation, I get an exit velocity of 26.48m/s after a distance of 50m.

Sub in ρ, k and m [Figure 8]

$$v = \sqrt{\frac{2177.82 e^{0.0013725} - 2177.82 + 0.2058(u)^2}{0.2058 e^{0.0013725}}}$$

Sub in distance of 50m

$$v = 26.48 \text{ m/s}$$

Before After Energy Loss

$$mgh = \frac{1}{2}mv^2 + E_{\text{Lost}}$$

$$300(9.81)(40) = \frac{1}{2}(300)(26.48)^2 + E_{\text{Lost}}$$

$$E_{\text{Lost}} = 12541.44 \text{ Joules}$$

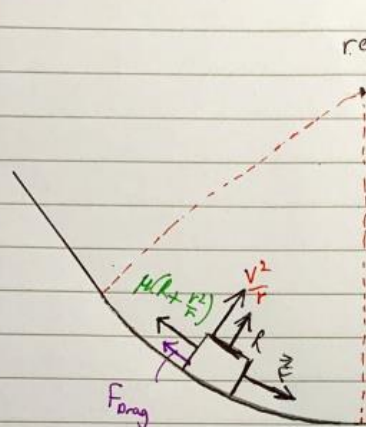
For the curve section of the third iteration, all I had to change was the acceleration of my recursive function to factor in drag. This gave me an acceleration of

$$a = g \sin(\tan^{-1}(\text{slope})) - \mu \left(g \cos(\tan^{-1}(\text{slope})) + \frac{v^2}{r} \right) - \frac{\frac{1}{2} \rho v^2 C_d A}{m}$$

ITERATION 3: Curve - Dynamic recursive [Figure 9]

$$F_d = \frac{1}{2} \rho v^2 C_d A$$

resultant $\vec{F} = \vec{F} - \mu \left(R + \frac{mv^2}{R} \right) - \frac{1}{2} \rho v^2 C_d A$

$$a = \frac{\vec{F} - \mu \left(R + \frac{mv^2}{R} \right) - \frac{1}{2} \rho v^2 C_d A}{m}$$


$$a = g \sin(\tan^{-1}(\text{slope})) - \mu \left(g \cos(\tan^{-1}(\text{slope})) + \frac{v^2}{15} \right) - 0.000686 v^2$$


```

public static void ITERATION2()
{
    double v = 26.94;
    Console.WriteLine(v.ToString());
    for (int n = 1; n < alpha; n++)
    {
        v = Math.Sqrt((v * v) +
            2 * (
                g * Math.Sin(
                    Math.Atan((15 * Math.Cos(T(n)) - 15 * Math.Cos(T(n + 1))) / (15 * Math.Sin(T(n)) - 15 * Math.Sin(T(n + 1))))
                )
            - mu * (g * Math.Cos(Math.Atan((15 * Math.Cos(T(n)) - 15 * Math.Cos(T(n + 1))) / (15 * Math.Sin(T(n)) - 15 * Math.Sin(T(n + 1))))
            ) + (v * v) / 15))
            *
            Math.Sqrt(((15 * Math.Cos(T(n + 1)) + 15 * Math.Cos(T(n))) * (-15 * Math.Cos(T(n + 1)) + 15 * Math.Cos(T(n))))
            + ((15 * Math.Sin(T(n + 1)) + 15 * Math.Sin(T(n))) * (-15 * Math.Sin(T(n + 1)) + 15 * Math.Sin(T(n))))
            ));
        Console.WriteLine(v.ToString());
    }
}

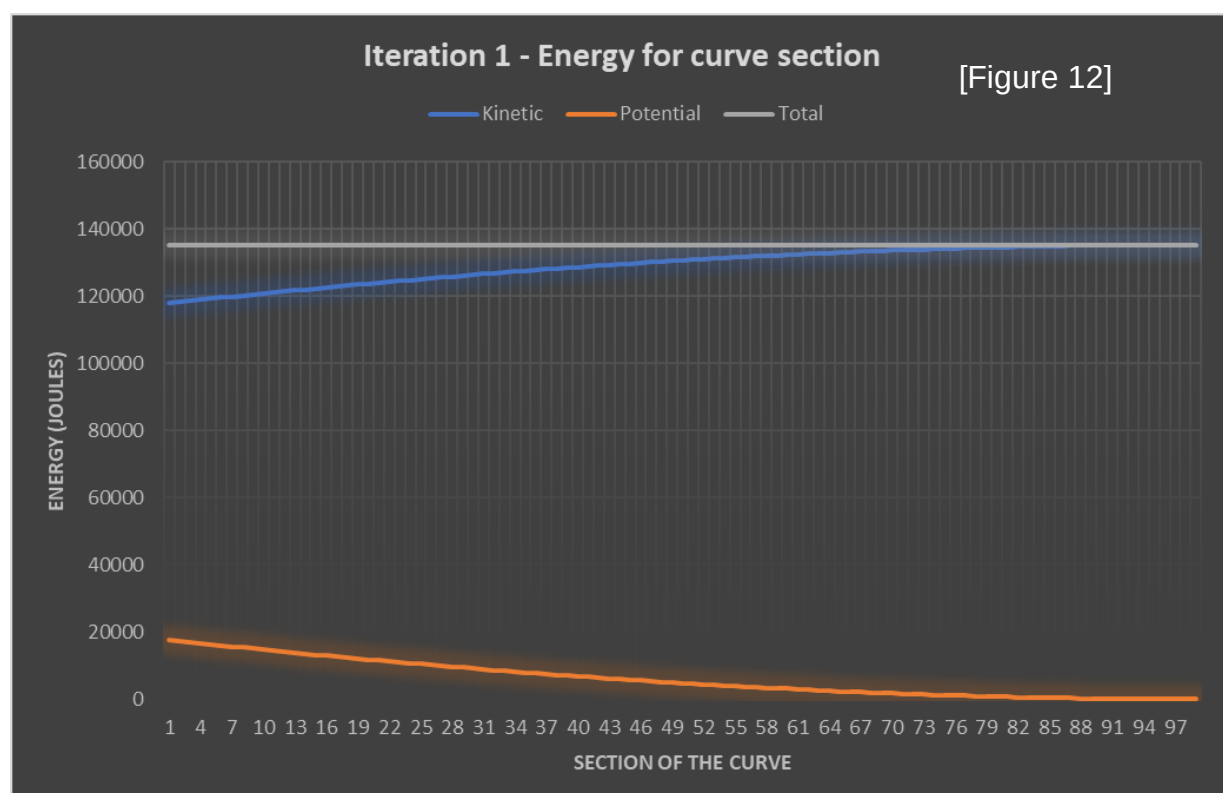
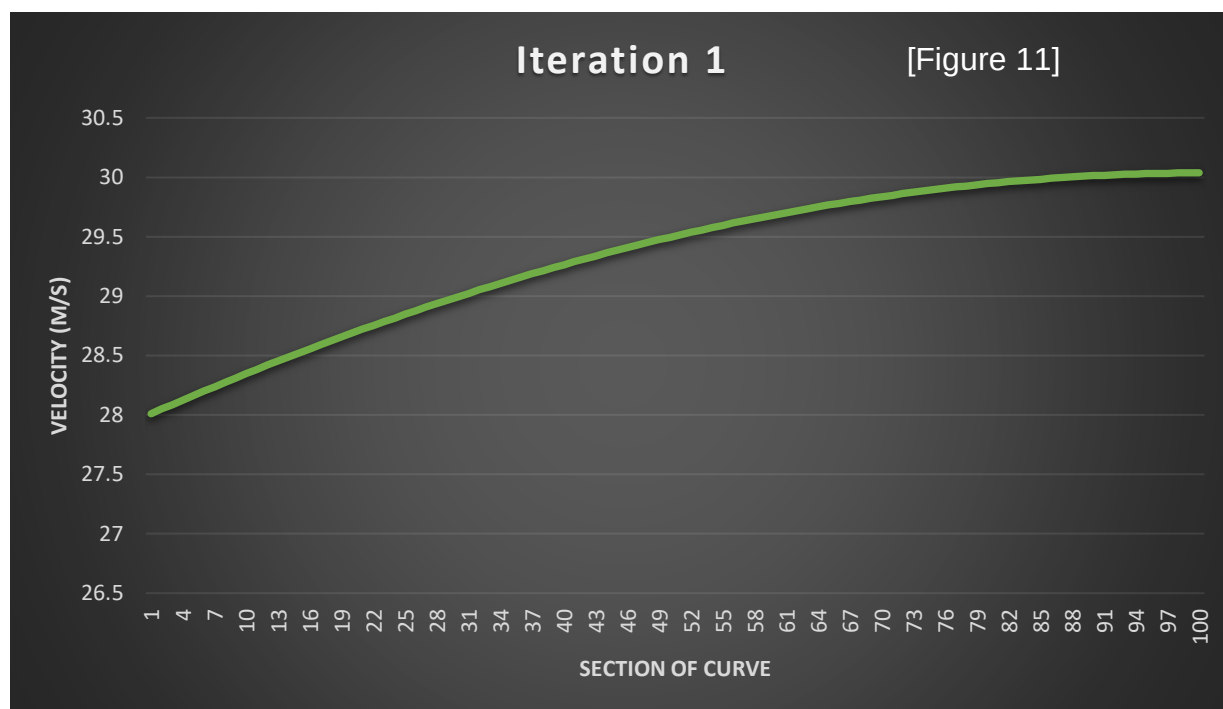
1 reference
public static void ITERATION3()
{
    double v = 26.48;
    Console.WriteLine(v.ToString());
    for (int n = 1; n < alpha; n++)
    {
        v = Math.Sqrt((v * v) +
            2 * (
                g * Math.Sin(
                    Math.Atan((15 * Math.Cos(T(n)) - 15 * Math.Cos(T(n + 1))) / (15 * Math.Sin(T(n)) - 15 * Math.Sin(T(n + 1))))
                )
            - mu * (g * Math.Cos(Math.Atan((15 * Math.Cos(T(n)) - 15 * Math.Cos(T(n + 1))) / (15 * Math.Sin(T(n)) - 15 * Math.Sin(T(n + 1))))
            ) + (v * v) / 15 - DragAcc(v))
            *
            Math.Sqrt(((15 * Math.Cos(T(n + 1)) + 15 * Math.Cos(T(n))) * (-15 * Math.Cos(T(n + 1)) + 15 * Math.Cos(T(n))))
            + ((15 * Math.Sin(T(n + 1)) + 15 * Math.Sin(T(n))) * (-15 * Math.Sin(T(n + 1)) + 15 * Math.Sin(T(n))))
            ));
        Console.WriteLine(v.ToString());
    }
}

```

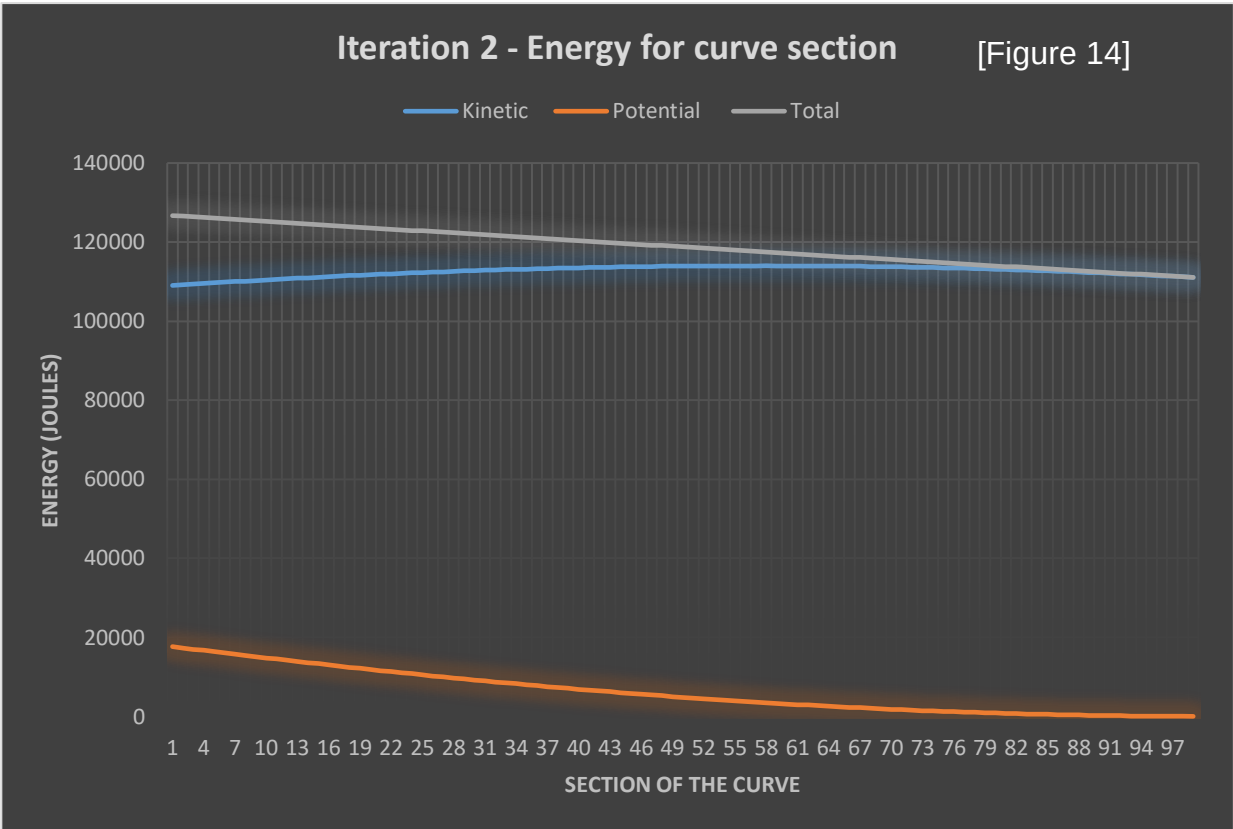
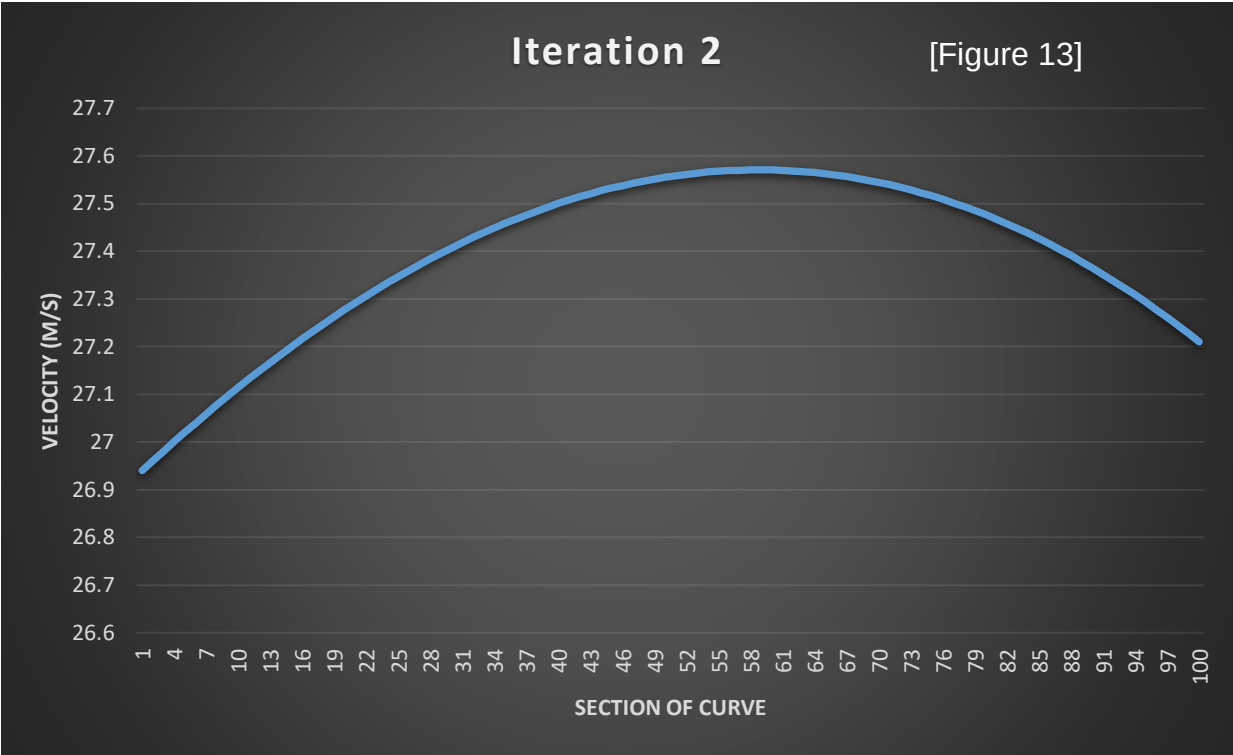
[Figure 10]

Figure 10 shows the code I wrote to generate the values for my velocity graphs.

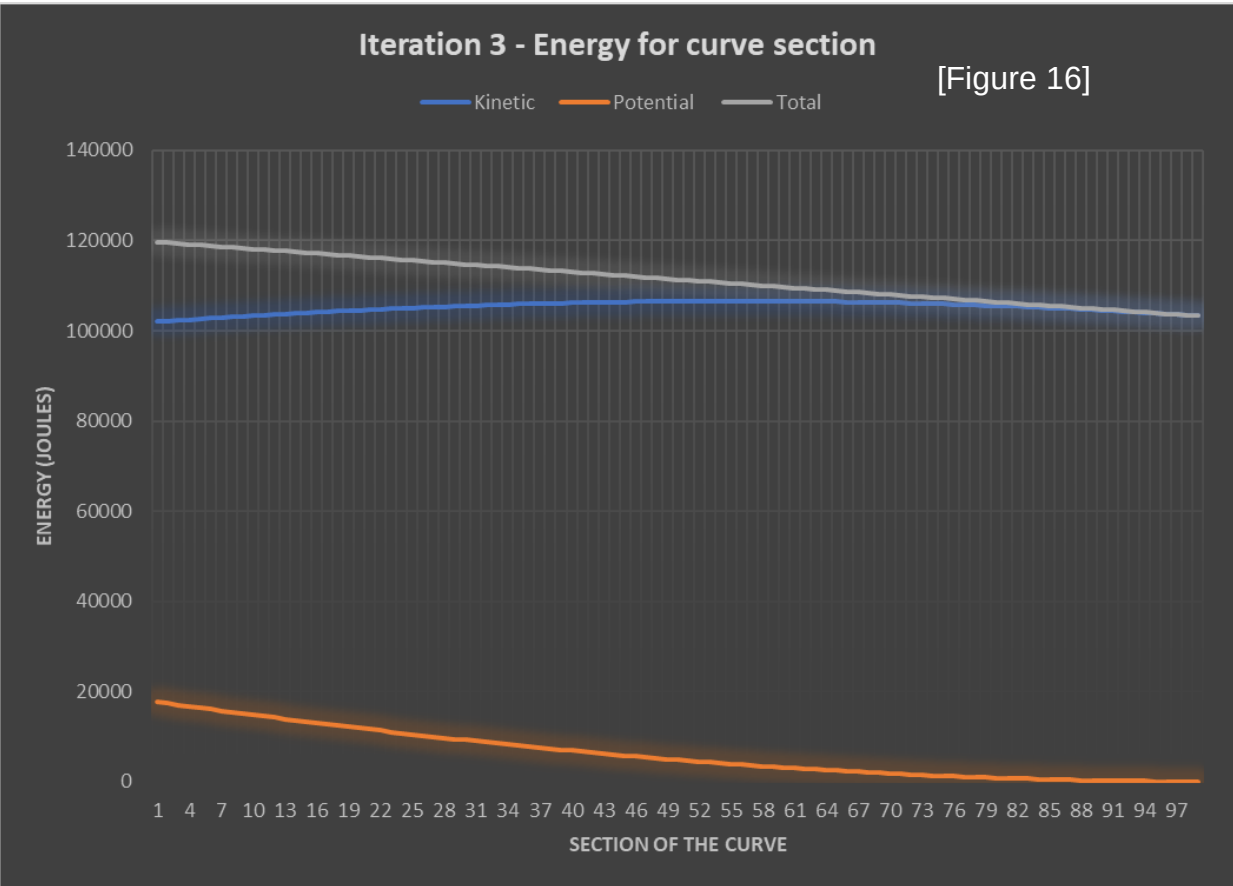
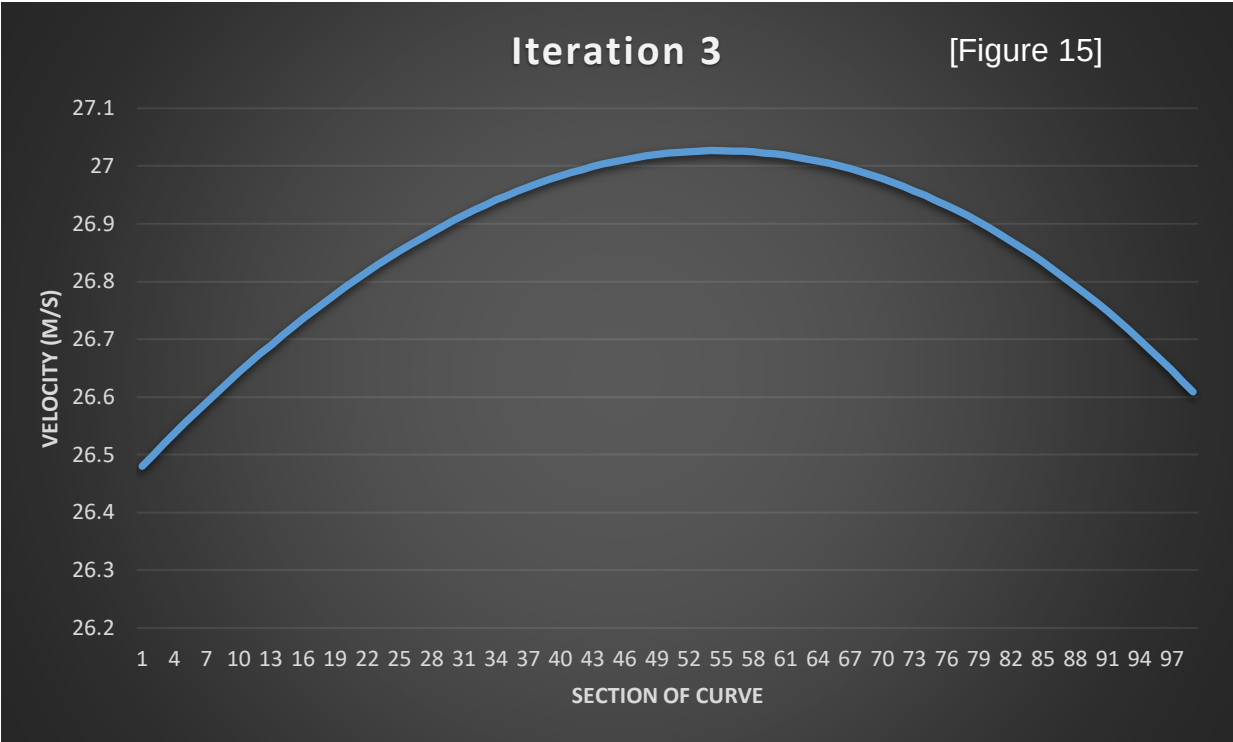
Iteration 1 was very simple assuming no drag, friction or loss of energy. Figure 11 is a velocity graph for the curve section. Figure 12 is an energy graph, note that the total energy is constant.



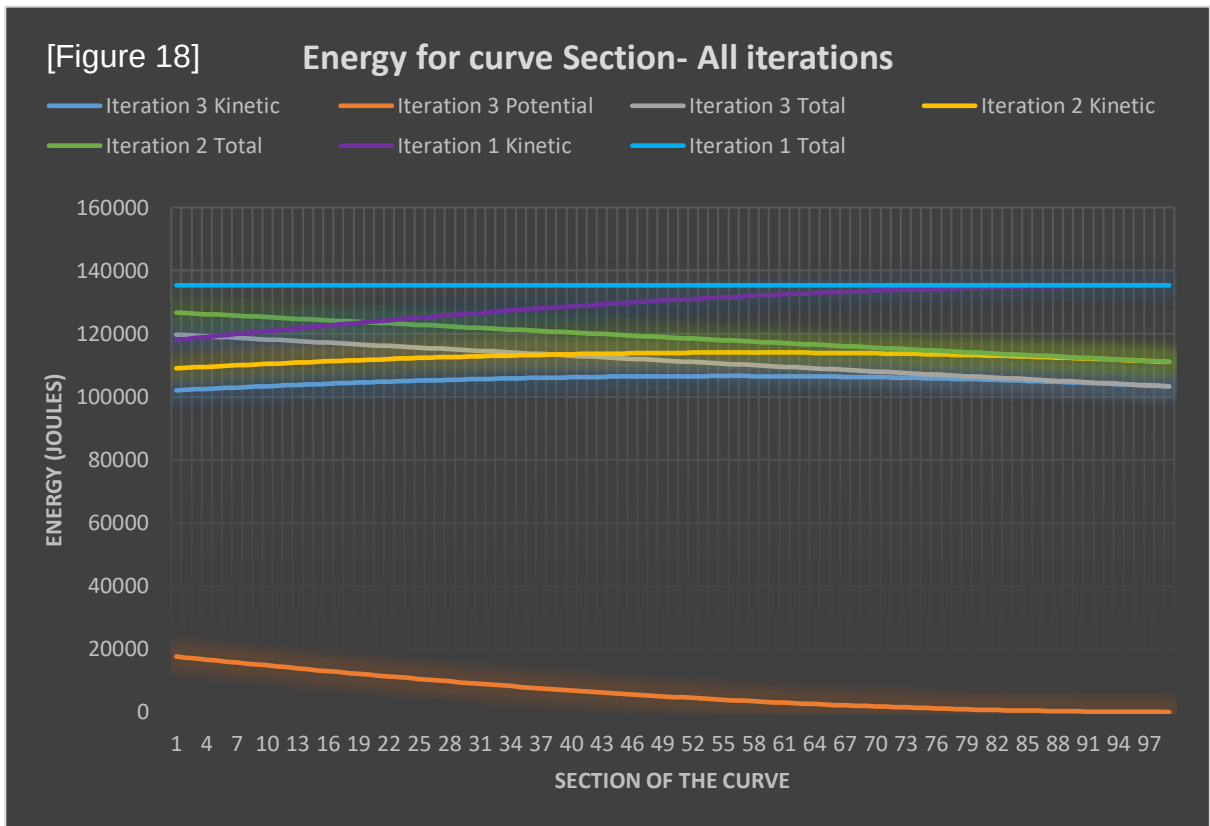
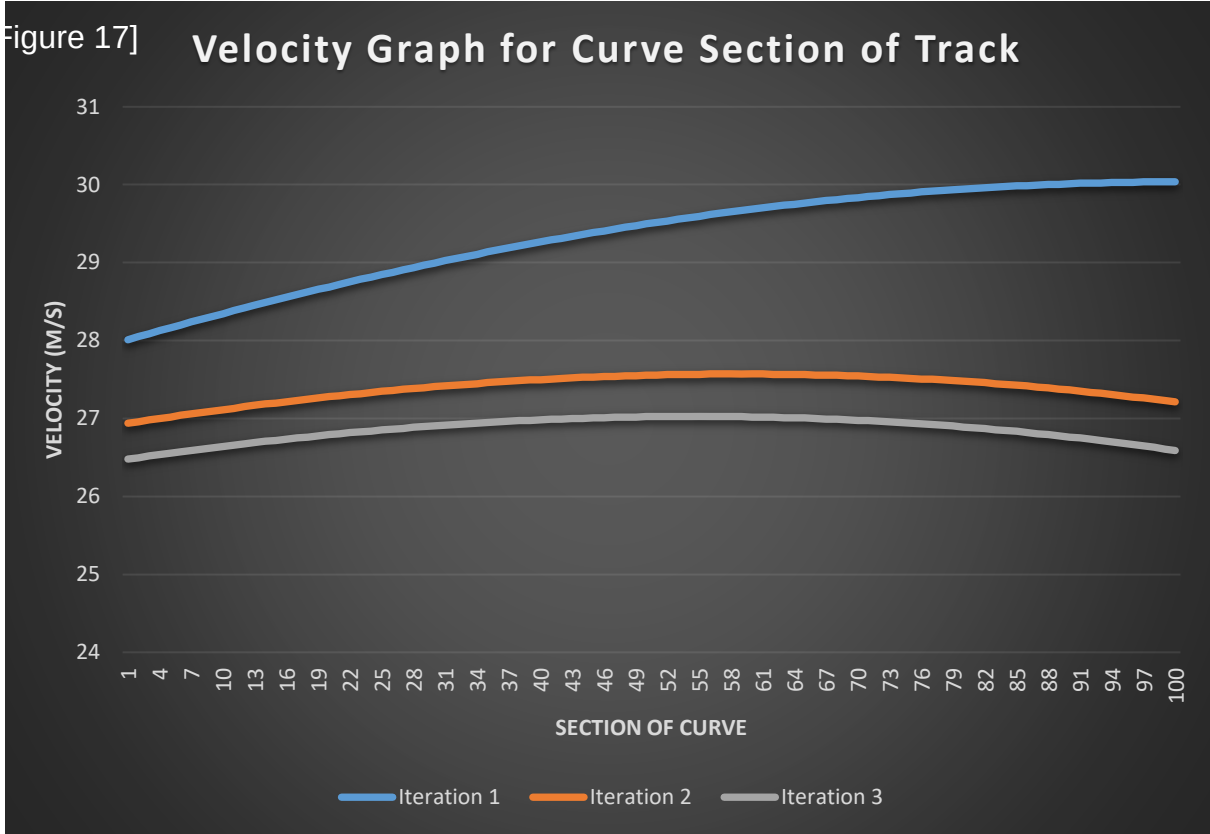
In Iteration 2, I assumed a coefficient of friction of 0.1 and created my approximation by a recursive function. Figure 13 is a velocity graph. Figure 14 is an energy graph, note that total energy declines.



In Iteration 3, I added on to my previous approximation by factoring in drag to the acceleration. Figure 15 is a velocity graph. Figure 16 is an energy graph, note that total energy declines steeper than figure 14.



Here are the 3 Iterations in comparison.



Conclusion

It is clear from my comparative graphs that over the course of my 3 iterations the result gets more accurate. Iteration 3 is the most accurate, however it is not real world. It is still an approximation, not an exact solution. My approximation doesn't factor in wind, air density changes, air temperature, etc. In further iterations I could factor in the weight and shape of the passengers, the drag caused by passengers or the exact solution to my approximation.